

Experiment Title

Division of Two Numbers Using 8086 Assembly Language

Objective

The purpose of this lab is to perform division between two numbers using 8086 assembly language and to analyze how quotient and remainder are stored after execution.

Required Tools

- Intel 8086 Architecture
- Assembly Language
- Assembly Language Emulator
- Personal Computer

Theory

The DIV instruction in the 8086 microprocessor is used for unsigned division. Before division, the dividend must be placed in the AX register. After execution, the **quotient** is stored in AL, while the **remainder** is stored in AH. Clearing AH before division is necessary to avoid incorrect results.

Procedure

1. Initialize the data segment.
2. Load the dividend into the AX register.
3. Load the divisor into a general-purpose register.
4. Execute the division instruction.
5. Observe quotient and remainder in registers.

Program Code

```
.model small
.stack 100h
.data
    dividend db 20
    divisor db 2
    msg db "Quotient = $"
.code
main proc
    mov ax, @data
    mov ds, ax
```

; division

mov al, dividend

mov ah, 0

mov bl, divisor

div bl ; AL = quotient

mov bx, ax ; save quotient

; print message

mov dx, offset msg

mov ah, 09h

int 21h

; convert quotient to decimal

mov ax, bx

mov cx, 0

mov bx, 10

convert2:

xor dx, dx

div bx

push dx

inc cx

test ax, ax

jnz convert2

print2:

pop dx

add dl, 30h

```
mov ah, 02h
```

```
int 21h
```

```
loop print2
```

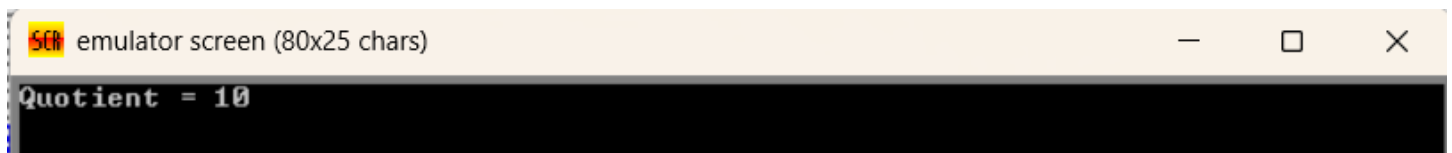
```
mov ah, 4ch
```

```
int 21h
```

```
main endp
```

```
end main
```

Output

A screenshot of a software emulator window titled "SCM emulator screen (80x25 chars)". The window has a standard Windows-style title bar with minimize, maximize, and close buttons. The main area of the window is black, and the text "Quotient = 10" is displayed in a green, monospaced font.

The division is completed successfully. The quotient is stored in AL and the remainder is stored in AH.

Conclusion

Through this experiment, the division mechanism of the 8086 processor was clearly understood. The lab emphasized correct register preparation and demonstrated how arithmetic results are distributed across registers.